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Homework 7

October 2, 2026

Problem 2)

Paper 1: A map of the day-night contrast of the extrasolar planet HD 189733b

This paper focuses on determining whether the atmosphere of HD 189733B, a hot Jupitar extrasolar planet, is able to transport energy to the nightside. When measuring the dayside using the IRAC on the Spitzer Space Telescope, there was a systematic error of a detector induced ramp. It was determined the ramp depended on illumination: in bright pixels, the fake signal climbs fast and stabilizes while lower flux pixels increase linearly over time. This error was corrected by taking the median illumination at each pixel and using interpolation with the natural logarithm of illumination. After the correction of this error, there was an observed change in brightness between the transit and secondary eclipse. After fitting data, analyzing the amplitude, and modeling the star along with its star spots as black bodies, it was determined that starspots could cause 0.1% linear increase in measured flux while the planetary phase variation was only 0.12%. After further investigation, it was concluded that phase variation was a planetary signal while recognizing possible contamination from starspots.

Additionally, there was a time delay when measuring the Secondary eclipse by 120 seconds. Scientists divided the planet into twelve slices, so each slice was 30 degrees longitude to create a rough brightness map of the planet at a wavelength of eight microns. This map revealed the planet is not uniformly bright; the brightest/hottest region was shifted eastward versus the expected hottest region directly beneath the star. This brightness induced a timing error that could explain at most, a 20 second delay, which does not fully explain the observed delay. Finally, when analyzing the atmospheric model and the shifted brightness maximum, it can be interpreted that winds can move a substantial amount if heat toward the night side before the atmosphere cools. The amount of opacity's effect on the radiative timescale is also taken into account. Overall, hot Jupiter atmospheres may have two important properties: they absorb most of the incoming visible starlight, and they efficiently transport much of that absorbed energy to the night side.

Paper 2: The Broadband Infrared Emission Spectrum of the Exoplanet HD 189733b

This paper focuses on measuring the infrared emission spectrum of the hot Jupiter HD 189733b to better understand its atmosphere. The secondary eclipse was observed through the Spitzer

Space Telescope's IRAC and MIPS instruments at different infrared wavelengths. Measuring the flux of the planet before the eclipse and the flux of the star during the eclipse allows researchers to estimate the planet's infrared emission. Cycling between the 4 IRAC channels led to obtaining 5-micron band measurements simultaneously. Since the shortest wavelengths channels use InSb and the longer wavelength channels use Si:As, the analysis of their data had to be done separately. Data processing of both channels included aperture photometry, background subtraction, fitting data, determining correction parameters using MCMC, and outlier rejection. Similar processing was used from the 24-micron wavelength data obtained from the MIPS instrument. The researchers then fitted the eclipse models to determine the depths and the planet's emission at each wavelength.

It was found that HD 189733b's infrared spectrum is different from a blackbody model due to the evidence of atmospheric molecules, particularly water vapor. The planet was then compared with HD 209458b, which had different atmospheric temperature structures, indicating that not all Hot Jupiter exoplanets have similar atmospheric properties. Further analysis in data suggested HD 189733b's atmospheric circulation may vary with altitude; different wavelengths can reveal different day to night temperature contrasts. The bandpasses also indicated that the planet may have a circular orbit. Overall, infrared observations, once thoroughly analyzed and fitted, can offer perspective of an exoplanet's atmospheric structure and how hot Jupiter's are not a uniform group.

Paper 3: The Deep Blue Color of HD 189733b

This paper focuses on the secondary eclipse observation for HD 18733b across the wavelength range 290-570nm made using the STIS on Hubble, which will help to measure the decrease in brightness when the planet's light disappears behind its star. The observations were divided into different wavelength ranges; however, the measurements of the light curve contained systematic variations caused by Hubble's thermal cycle and the movement of the spectrum across its detector. A Gaussian Processes was used to model these effects simultaneously with the planetary eclipse, and MCMC sampling was applied to estimate the eclipse's depths and uncertainties. Comparing the eclipse depths at different wavelengths indicated HD reflected more blue light than longer-wavelength visible light.

Further analysis was done by creating an atmospheric model to study reflective clouds, absorption by sodium, and Rayleigh scattering. It was found that optically thick reflective clouds above absorption maintain high albedo measurements, but no clouds/optically thick clouds under absorption can lower the albedo. This is further supported by the finding of how sodium absorption reduces the reflected light at longer wavelengths while the Rayleigh Scattering helps redirect shorter wavelengths toward space; the combination of these two elements produces a blue reflection spectrum. The researchers also suggested that the high-altitude clouds contain

very small particles which could help explain why the planet reflects blue light effectively, but they could not determine the definite cloud composition or altitude. Overall, secondary eclipse observations can provide insight into the atmospheric properties of exoplanets.

Paper 4: Retrievals on NIRCam transmission and emission spectra of HD 189733b with PLATON 6, a GPU code for the JWST era

Researchers in this paper used JWST'S NIRCAM to examine wavelengths between 2.4 and 5 microns to observe both the transit and secondary eclipse of HD 18733b to determine the atmospheric composition. SPARAT was used for data reduction of the raw telescope data to process observations and correct instrumental effects to measure how the observed light changes with wavelength and time. To analyze the spectra, an improved software called PLATON 6 was used. This program generates atmospheric models, has GPU acceleration, and additional statistical tools to help evaluate the strength of evidence. Comparing the created atmospheric models with the observed spectra will help determine the atmospheric properties of HD 18733b; this paper also aims to evaluate their retrieval methods.

The results showed a detection of 4 atmospheric molecules present in both transmission and emission spectra: Water Vapor, Carbon Dioxide, Carbon Monoxide, and Hydrogen Sulfide. This data indicated that HD 18733b's atmosphere has a metallicity approximately 2.6 - 6.8 times that of the sun and has a carbon-to-oxygen ratio of 0.42. Additionally, there seems to be a methane depletion near the planet's terminator which, according to VULCAN, could be due to photochemistry. An investigation of HD 18733b's temperature-pressure profile found evidence of a dayside thermal inversion at low pressures but at a low statistical significance. These results indicate that the planet has a complex atmosphere; this is further supported by the observation that assumptions about atmospheric chemistry can be more critical than cloud-model choices when determining HD 18733b's composition.